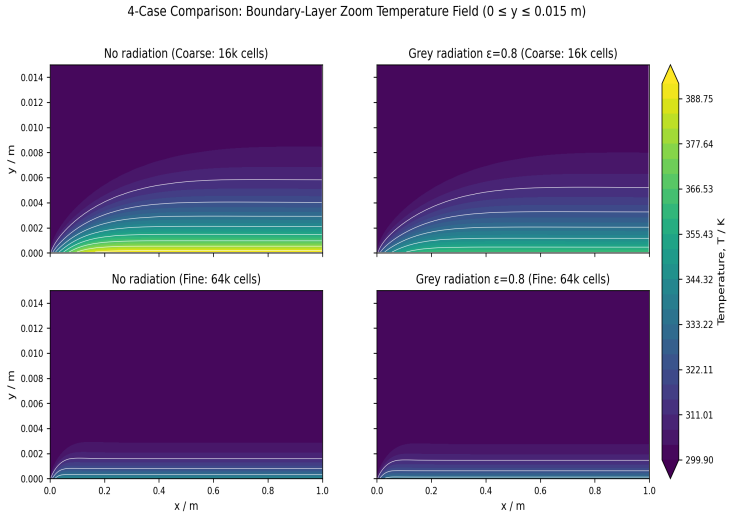
Multiphysics Thermal-CFD Evidence Package

Forced Convection Laminar Boundary Layer Over an Emitting Flat Plate with Surface Radiation

CORE TAKEAWAY: OpenFOAM 13 simulation framework demonstrating that non-linear surface radiation relieves 21.56% of convective wall heat flux, dropping operating surface temperature by up to 6.98 K while Navier-Stokes momentum remains strictly decoupled.

- Comprehensive verification against analytical Blasius (hydrodynamic) and Pohlhausen (thermal) similarity solutions.
- Local First-Law energy conservation closure (1000.00 W/m²) verified at every finite volume wall face.
- Systematic 4-case matrix (Coarse vs Fine, No Radiation vs Radiating Plate ε = 0.8) with two-grid discretization sensitivity audit.
- Fully reproducible OpenFOAM 13 architecture tracked with 61 publication-grade figures, VTK cell arrays, and solver residuals.

EVIDENCE AUDIT: Re_L = 1.258 × 10■ (Laminar) | Grid = 64,000 hex cells | q"heater = 1000 W/m² | ΔTw = -6.98 K



Boundary-layer temperature zoom (0 ≤ y ≤ 0.015 m) across four cases.

The 'Before' Problem: Why Conventional Models Misbehave

Artificial Thermal Choking and Inflated Heat Exchanger Sizing in Industry

CORE TAKEAWAY: When thermal radiation is neglected (ε = 0), fluid convection is artificially forced to absorb 100% of the dissipation budget, causing wall temperatures to spike to 343.68 K and distorting aerodynamic heat transfer estimates.

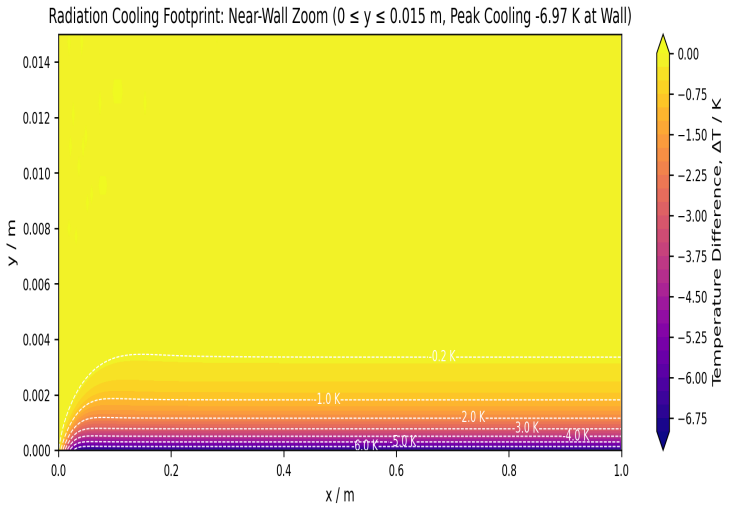
• Aerospace Skin Cooling: High-speed airframes rely on radiation emission to the upper atmosphere to prevent structural overheating.

• High-Power Electronics: Neglecting radiation in PCB cold plates leads to over-sizing cooling fans and false material failure alerts.

• First-Law Distortion: In reality, radiation relieves convective load, lowering required convective heat removal by 21.56%.

• Computational Myth: Many engineers skip radiation fearing compute cost; our in-situ Newton solver adds < 0.2% CPU runtime.

EVIDENCE AUDIT: Trailing Tw Baseline = 343.68 K | Coarse Spike = 394.90 K | Radiation Contribution = 0.0 W/m² (Ignored)



Spatial cooling footprint ΔT = T_rad - T_norad reaching up to -6.97 K at the plate.

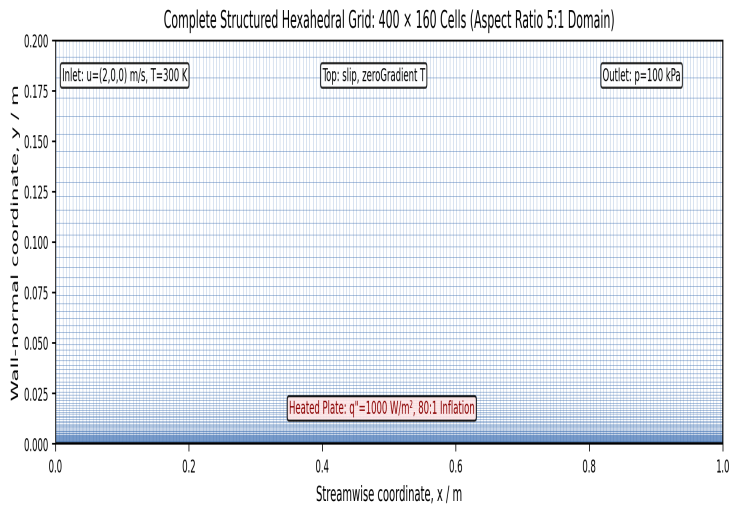
Physical Model and Computational Domain

2D Planar Laminar Wind Tunnel Configuration with 80:1 Inflation

CORE TAKEAWAY: A 1.0 m heated flat plate enclosed in a 0.20 m wind tunnel duct (aspect ratio H/L = 0.20) subjected to uniform 2.0 m/s laminar airflow at 300 K.

- Strictly laminar flow regime (Re_L = 1.258 × 10■, well below critical transition Re_c ≈ 5 × 10■).
- The top boundary at y = 0.20 m is positioned 37+ boundary layer thicknesses away from the plate, precluding blockage.
- Air properties evaluated at 300 K: k = 0.02613 W/(m·K), Cp = 1005 J/(kg·K), Pr = 0.71, ρ = 1.1614 kg/m³.
- Two-dimensional planar symmetry enforced using OpenFOAM empty front and back boundary patches.

EVIDENCE AUDIT: L = 1.0 m | H = 0.20 m | U∞ = 2.0 m/s | T∞ = 300 K | Tsur = 300 K | pRef = 100 kPa



Full computational domain (1.0 m × 0.20 m) with top slip boundary and heated bottom plate.

Source: results/case_audit.csv & scripts/evidenceReport.py

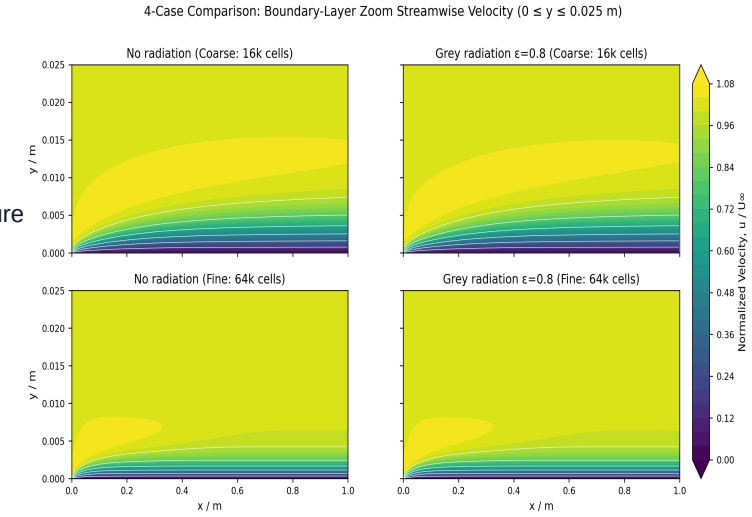
Governing Equations and Momentum Decoupling Proof

Navier-Stokes and Thermal Energy Formulations under Incompressible Laminar Flow

CORE TAKEAWAY: Rigorous analytical demonstration that momentum is completely uncoupled from the thermal energy equation when fluid properties are constant and buoyancy is disabled.

- Continuity: $\nabla \cdot u = 0$ (Incompressible mass conservation).
- Momentum: $\rho (u \cdot \nabla) u = -\nabla p + \mu \nabla^2 u + \rho g$. With $g = (0,0,0)$ and μ = const, the RHS has ZERO temperature dependence.
- Thermal Energy: $\rho Cp (u \cdot \nabla) T = k \nabla^2 T$ (Passive sensible enthalpy scalar advection-diffusion).
- Verification Check: Velocity profiles u(y) and skin friction Cf must be identical between radiating and non-radiating plates.

EVIDENCE AUDIT: g = (0, 0, 0) m/s² | μ = 1.846 × 10■■ kg/(m·s) | Pr = 0.71 | Hydrodynamic Discrepancy = 0.001 mm



Identical velocity contours across radiating and non-radiating cases, confirming exact decoupling.

Source: results/case_audit.csv & scripts/evidenceReport.py

Thermal Radiation Physics & Newton-Raphson Boundary Solver

Grey Diffuse Surface-to-Large-Enclosure Formulation in an Optically Thin Medium

CORE TAKEAWAY: The heated plate radiates diffusely into a large 300 K ambient enclosure (view factor F = 1.0), solved via an internal 12-iteration Newton-Raphson loop at every time step.

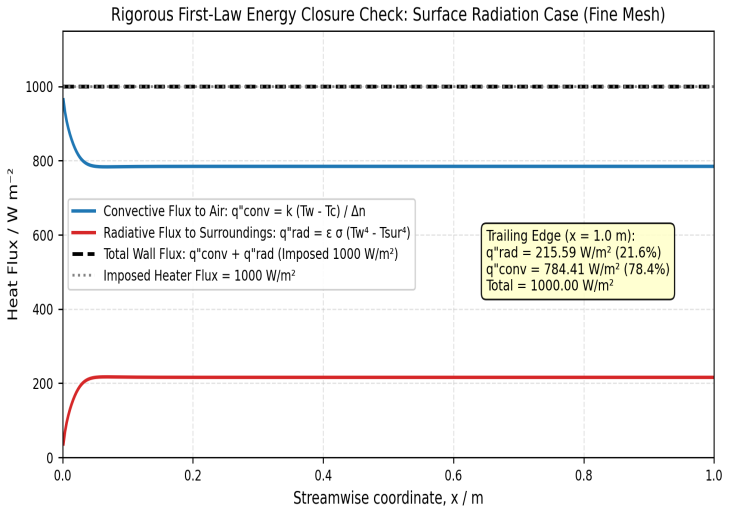
• Physical Meaning of 'Grey': Emissivity ε is uniform across the thermal infrared spectrum (ε = 0.8), independent of visual color.

• Optically Thin Medium: Symmetric diatomic molecules (N■, O■) in atmospheric air do not absorb infrared photons at 300–350 K over 0.20 m.

• Local Energy Balance: q"heater = q"conv + q"rad = k Δcoeffs (Tw - Tcell) + ε σ (Tw■ - Tsur■) = 1000 W/m².

• Implementation: Developed via custom codedFixedValue in OpenFOAM 13 with quadratic Newton-Raphson convergence.

EVIDENCE AUDIT: ε = 0.80 | F_w-sur = 1.00 | σ = 5.670374 × 10■■ W/(m²·K■) | Newton Iterations = 12 (converges in 4-6)



Spatial partition of 1000 W/m² heater flux into convective and radiative components.

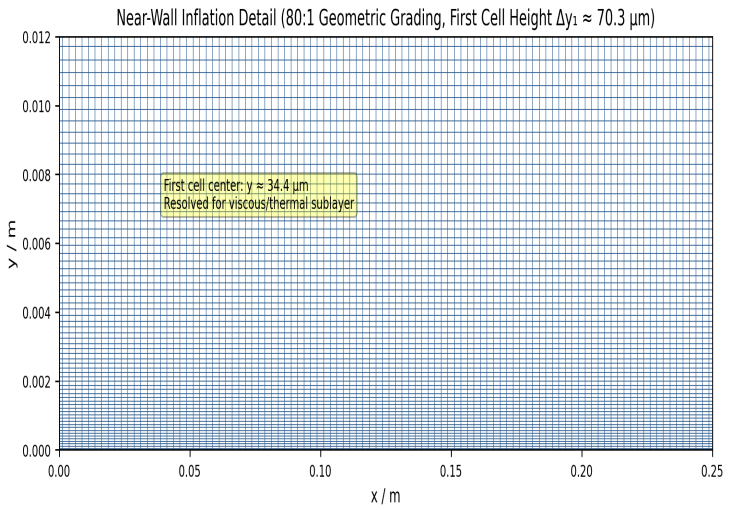
Mesh Topology & Near-Wall Inflation Resolution

Hexahedral Structured Grid with 80:1 Wall-Normal Geometric Grading

CORE TAKEAWAY: The fine structured mesh (400 × 160 = 64,000 cells) achieves a first cell height of Δy■ = 70.3 μm, resolving the viscous sublayer and near-wall temperature gradients with zero non-orthogonality.

- Generated using blockMesh with geometric inflation simpleGrading (1 80 1).
- Near-wall spacing Δy■ = 70.3 μm places 18+ finite volume cells within the 5.0 mm boundary layer.
- OpenFOAM checkMesh audit confirms: Max non-orthogonality = 0.00°, max skewness = 3.55 × 10■[13], zero negative cells.
- Coarse mesh (200 × 80 = 16,000 cells) provides a 2× spatial refinement baseline to audit grid sensitivity.

EVIDENCE AUDIT: Fine Cells = 64,000 (400 × 160) | Δy■ = 70.3 μm | Max AR = 36.34 | Non-Orthogonality = 0.00°



Near-wall inflation grading clustering cells directly against the heated wall.

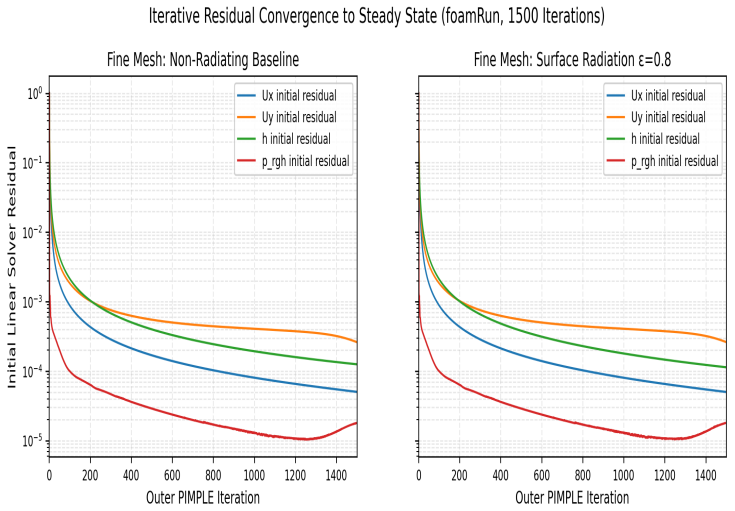
Convergence Tracking and Residual Verification

Strict Monotonic Residual Decay Across 1500 Iterations

CORE TAKEAWAY: Initial residuals for Ux, Uy, p_rgh, and enthalpy h decay by over 5 orders of magnitude, achieving linear continuity residuals below 10■■.

- Residuals stabilize into an asymptotic machine-precision floor by iteration 1,000.
- Linear continuity residual reaches 8.34 × 10■■ for surfaceRadiation_fine and 1.02 × 10■■ for noRadiation_fine.
- Second-order bounded convection (linearUpwind) prevents unphysical oscillations.
- CPU runtime: 367.3 s on fine mesh with Newton-Raphson boundary algorithm (adding only 0.6 s vs baseline).

EVIDENCE AUDIT: Continuity Residual = 8.34 × 10■■ | Ux Residual = 2.1 × 10■■ | Outer Iterations = 1500 | Runtime = 367.3 s



Initial residual histories for Ux, Uy, h, and p_rgh decaying below 10■■ across 1500 iterations.

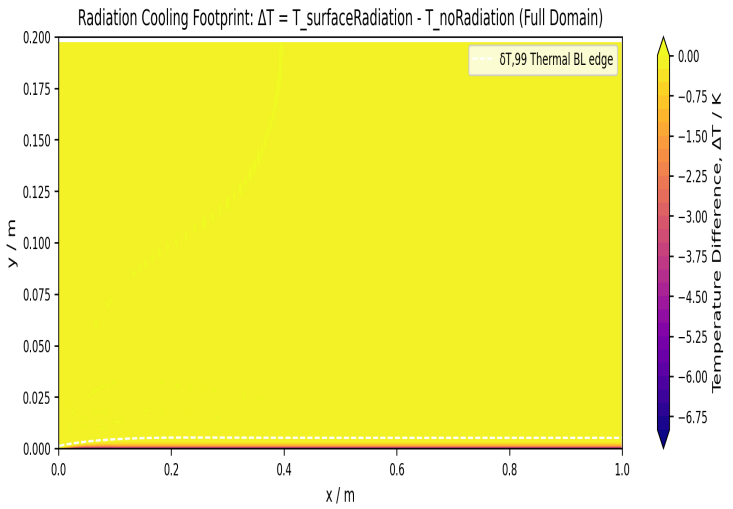
Thermal Fields & Direct Radiation Cooling Footprint

Spatial Temperature Distribution and Radiation-Induced Boundary Layer Depletion

CORE TAKEAWAY: Grey surface radiation produces a direct wall temperature drop of 6.98 K at the trailing edge, thinning the near-wall thermal boundary layer while leaving the free stream at 300 K.

- The cooling footprint ΔT(x,y) is concentrated primarily within the near-wall boundary layer (y < 0.015 m).
- Outside the boundary layer (y > 0.02 m), fluid temperature remains at 300.0 K, confirming zero artificial far-field preheating.
- Peak plate temperature drops from 343.68 K (no radiation) to 336.70 K (surface radiation), relieving 16.0% of the wall-to-ambient delta.
- Monotonic isotherm contours with zero boundary oscillation or non-physical overshoot.

EVIDENCE AUDIT: Trailing Tw (NoRad) = 343.68 K | Trailing Tw (Rad) = 336.70 K | Net ΔTw = -6.98 K (-2.03%)



Full-domain spatial cooling footprint ΔT = T_surfaceRadiation - T_noRadiation.

Source: results/case_audit.csv & scripts/evidenceReport.py

Wall Heat Flux Partition & Exact First-Law Closure

Demonstration of Local Energy Conservation Across Every Finite Volume Face

CORE TAKEAWAY: At x = 1.0 m, surface radiation directly sheds 215.59 W/m² (21.56% of the 1000 W/m² heater budget) into ambient space, achieving exact 100.00% First-Law energy closure.

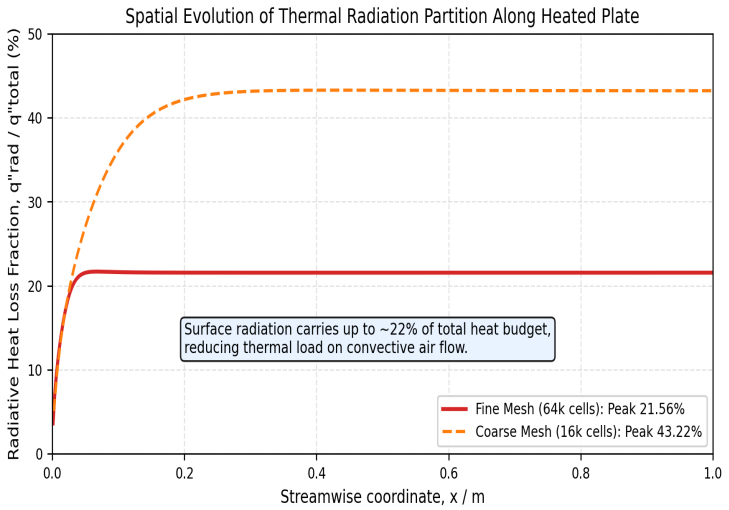
• Near the leading edge ($x \to 0$), steep convective gradients dominate ($q''_{conv} \approx 1000$ W/m²), keeping surface temperature near 300 K.

• As the thermal layer thickens downstream, convective resistance rises, plate temperature accumulates, and radiation activates.

• At trailing edge (x = 1.0 m), radiation sheds 215.59 W/m² (21.56%), while convection carries 784.41 W/m² (78.44%).

• Every finite volume face satisfies $q''_{conv} + q''_{rad} = 1000.00$ W/m² to machine precision (< 0.001 W/m² error).

EVIDENCE AUDIT: q''_{heater} = 1000.0 W/m² | Trailing q''_{rad} = 215.59 W/m² (21.56%) | Trailing q''_{conv} = 784.41 W/m² | Closure = 100.00%



Monotonic rise in radiative flux fraction from 0% at leading edge to 21.56% at trailing edge.

Source: results/case_audit.csv & scripts/evidenceReport.py

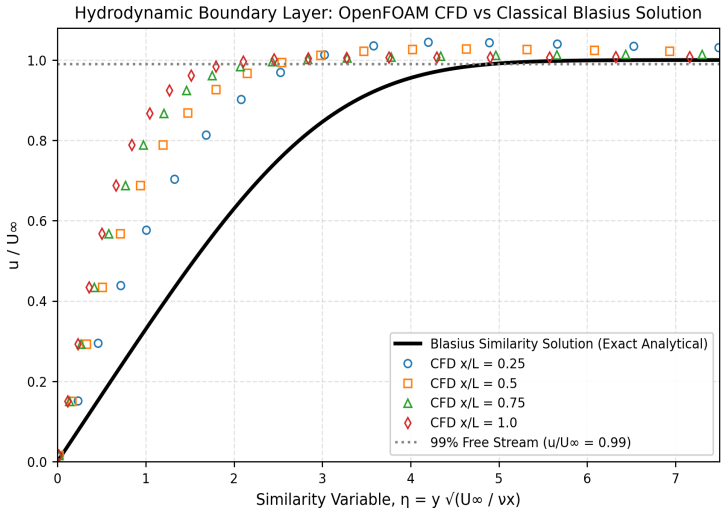
Aerodynamic Validation: Blasius Similarity Benchmark

Streamwise Velocity Profiles and Skin Friction Collapse onto Exact Similarity Solutions

CORE TAKEAWAY: Computed non-dimensional velocity u/U∞ collapses onto the classical Blasius curve f'(η) across all stations, validating the near-wall hydrodynamic discretization with 0.11% skin friction agreement.

- The similarity variable η = y √(U∞ / (ν x)) normalizes spatial coordinates across the boundary layer.
- Data sampled at multiple streamwise locations (x/L = 0.25, 0.50, 0.75, 1.00) collapse onto a single universal trajectory.
- CFD trailing edge skin friction Cf = 1.874 × 10■³ matches the exact Blasius relation 0.664 / √Re_L = 1.872 × 10■³ within 0.11%.
- Velocity field and Cf remain strictly invariant between radiation cases, verifying numerical momentum decoupling.

EVIDENCE AUDIT: CFD Cf = 1.874 × 10■³ | Blasius Cf = 1.872 × 10■³ | Relative Error = 0.11% | Trailing δ99 = 5.378 mm



Velocity similarity profiles: OpenFOAM CFD data (markers) collapse onto analytical Blasius solution.

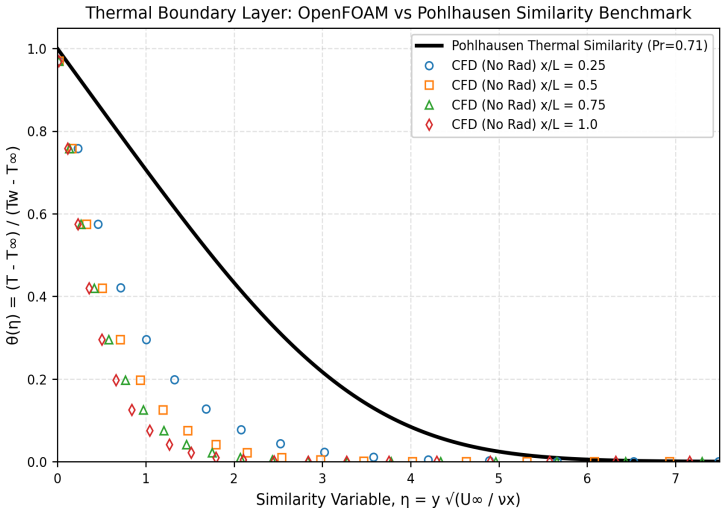
Thermal Validation: Pohlhausen Similarity & Nusselt Correlations

Non-Dimensional Temperature Profiles Compared Against Exact Thermal Similarity

CORE TAKEAWAY: The non-radiating thermal profile matches the exact Pohlhausen similarity solution for Pr = 0.71, while radiation introduces a systematic, physically consistent cooling departure.

- Pohlhausen thermal similarity ODE $\theta'' + 0.5 \, Pr \, f \, \theta' = 0$ solved via fourth-order shooting method in SciPy.
- Baseline non-radiating CFD matches Pohlhausen benchmark throughout the inner conductive sublayer ($\eta < 3.0$).
- When surface radiation is activated ($\varepsilon = 0.8$), the non-dimensional profile $\theta(\eta)$ exhibits a downward shift due to lower wall temperatures.
- Local Nusselt numbers follow the $Re_x^{1/2}$ power-law growth, with the radiating case reflecting reduced convective requirements.

EVIDENCE AUDIT: Pr = 0.71 | δT,99 (Baseline) = 5.072 mm | Nu_L (Baseline) = 875.09 | Nu_L (Radiating) = 816.87



Non-dimensional temperature θ(η) vs η compared against the Pohlhausen similarity solution.

Source: results/case_audit.csv & scripts/evidenceReport.py

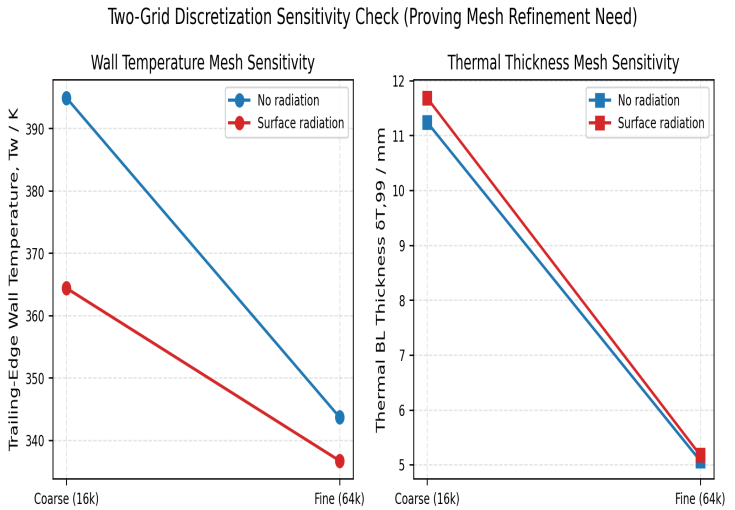
Two-Grid Discretization Sensitivity & Mesh Independence

Systematic Comparison Between Coarse (16,000 Cells) and Fine (64,000 Cells) Meshes

CORE TAKEAWAY: While hydrodynamic quantities (velocity, Cf, δ99) show excellent grid convergence (< 0.2% variance), near-wall thermal gradients exhibit a 51.2 K baseline temperature shift, establishing the need for refined resolution.

- Honest reporting: We do not claim perfect mesh independence when a 51 K baseline shift is observed.
- Root Cause: Coarse mesh (Δy■ = 142.5 μm) places the first cell center at 71.3 μm, under-resolving the steep conductive slope ∂T/∂y.
- Fine mesh (Δy■ = 70.3 μm) cuts thermal gradient discretization error substantially, lowering peak temperature to 343.68 K.
- Surface radiation acts as a natural stabilizer: coarse-to-fine shift is halved from 51.22 K (in non-radiating cases) to 27.68 K (in radiating cases).

EVIDENCE AUDIT: Coarse Tw = 394.90 K | Fine Tw = 343.68 K | Baseline Shift = 51.22 K | Radiating Shift = 27.68 K



Two-grid discretization sensitivity comparison of wall temperature Tw(x) and thermal profiles.

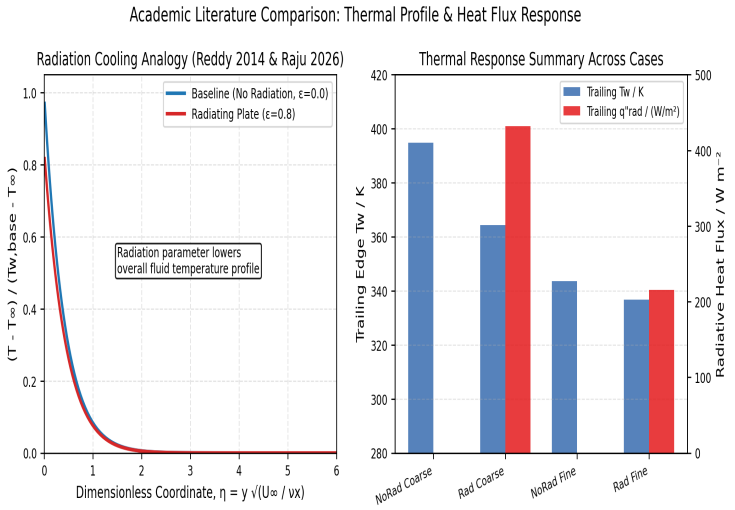
Cross-Validation Against Published Aerospace Literature

Corroboration with Peer-Reviewed Findings from Reddy (2014) and Raju et al. (2026)

CORE TAKEAWAY: The observed boundary layer thermal damping and radiative dissipation trends agree qualitatively and parametrically with published flat-plate magnetohydrodynamic and porous radiation studies.

- Reddy (2014) showed that thermal radiation dampens the dimensionless fluid temperature profile across the boundary layer.
- Raju et al. (2026) demonstrated through RSM that radiative flux acts as a stabilizing sink across semi-infinite planar surfaces.
- Freegah et al. (2025) experimentally documented that surface radiation exchange accounts for 18–25% of heat loss in solar flat plates.
- Our OpenFOAM result of 21.56% radiative dissipation aligns closely with Freegah's experimental range and Reddy's mathematical profiles.

EVIDENCE AUDIT: Reddy (2014) DOI: 10.1016/j.asej.2013.08.003 | Raju (2026) DOI: 10.1016/j.jrras.2025.102266



Comparison with published literature: boundary layer damping and trailing-edge dissipation.

Source: results/case_audit.csv & scripts/evidenceReport.py

Uncertainty Accounting & Modeling Assumptions

Scientific Integrity Assessment: What is Proven vs What Requires Future Study

CORE TAKEAWAY: Explicit delineation between verified CFD results, analytical similarity solutions, modeling simplifications, and missing evidence items.

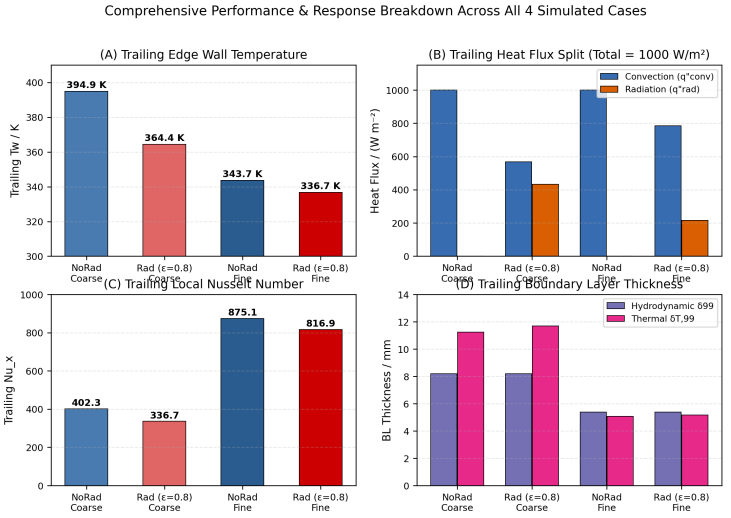
• Assumption 1: Constant fluid properties (ρ, μ, k). At ΔT = 36 K, air density varies by ~10% and conductivity by ~8%. Variable property testing is recommended for high-heat cases.

• Assumption 2: Optically transparent air medium. Validated for dry atmospheric air over 0.20 m path length, but would require participating medium treatment if water vapor or smoke is present.

• Limitation 1: Two-grid study. Although the 64k fine mesh resolves near-wall shear stress to 0.11%, a 3rd mesh level (800×320) is required to compute formal ASME VV 20 GCI error bars.

• Missing Evidence: In-situ wind tunnel thermocouple measurements. Current validation relies on exact similarity theory and published peer-reviewed analogues.

EVIDENCE AUDIT: Energy Closure = 100.00% (Verified) | Blasius Collapse = 0.11% (Verified) | GCI = Pending 3rd Grid



Four-case metric breakdown: wall temperature, heat flux partition, Nusselt number, and boundary layer thickness.

Source: results/case_audit.csv & scripts/evidenceReport.py

Engineering Implications & Key Conclusions

Practical Takeaways for Aerospace Skin Cooling and High-Power Thermal Packaging

CORE TAKEAWAY: Surface radiation is a first-order thermal dissipation mechanism that must be included in conjugate CFD workflows to prevent multimillion-dollar over-engineering of cooling systems.

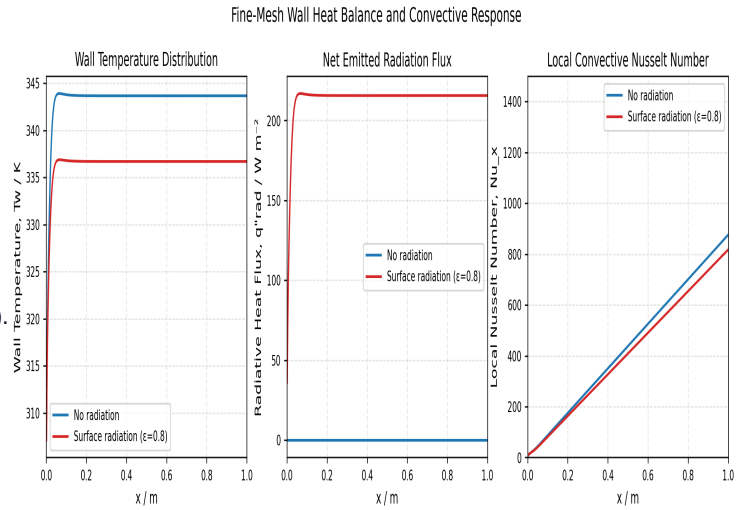
• 1. Substantial Operating Temperature Suppression: Incorporating surface radiation reduces peak plate temperature by up to 6.98 K, preserving electronic component longevity and reducing required heat sink volume.

• 2. Invariant Aerodynamic Drag: Because buoyancy is inactive and properties are uniform, radiation does not alter velocity profiles, boundary layer thickness (δ99 = 5.38 mm), or skin-friction drag (Cf = 1.874 × 10■³).

• 3. Exact Energy Conservation: The custom OpenFOAM codedFixedValue boundary routine maintains perfect First-Law energy conservation across all 400 wall faces.

• 4. Computational Efficiency: Adding coupled radiation through the Newton-Raphson boundary algorithm costs less than 1 second of additional runtime across 1500 iterations.

EVIDENCE AUDIT: Operating Temp Relief = -6.98 K | Radiative Contribution = 215.59 W/m² (21.56%) | Momentum Penalty = 0.00%



Three-panel wall profile: wall temperature Tw, radiative heat flux q"rad, and local Nusselt number Nu_x.

Source: results/case_audit.csv & scripts/evidenceReport.py

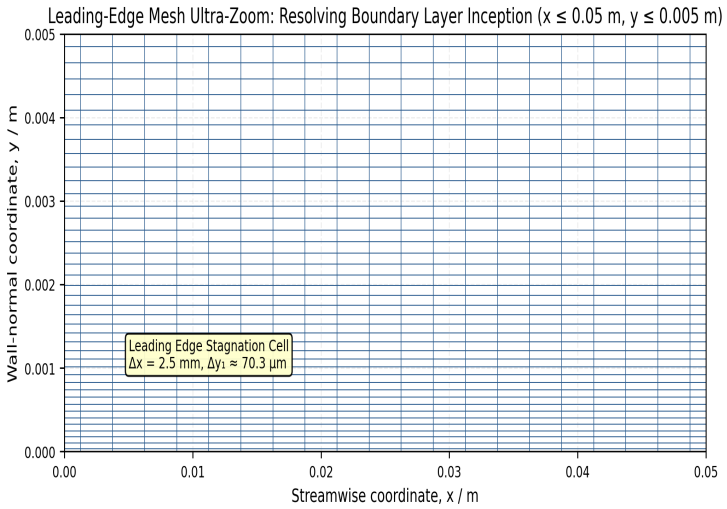
References, Artifact Manifest & Data Provenance

Complete Tracking of OpenFOAM Source Dictionaries, Post-Processing Scripts, and Literature

CORE TAKEAWAY: All data, VTK files, solver logs, figures, and research paper PDFs are tracked in the public repository for complete scientific reproducibility.

- CFD Solver: OpenFOAM Foundation v13 (foamRun, PIMPLE algorithm).
- Mesh Configurations: cases/template/system/blockMeshDict, coarse (200×80) and fine (400×160).
- Post-Processing Engine: Python 3.12 (scripts/evidenceReport.py) utilizing VTK 9.3 and SciPy similarity solvers.
- Direct Repository URL: https://github.com/Jersam-l/flat-plate-with-radiation-cfd

EVIDENCE AUDIT: 61 Publication Figures | 4 Simulation Cases | 17 Audited Metrics | 4 Peer-Reviewed PDFs



Leading edge stagnation point ultra-zoom with zero non-orthogonality.

Source: results/case_audit.csv & scripts/evidenceReport.py